CIS 5270 Midterm 1 - Rough Review Notes

Lecture 3 to Lecture 10 March 2026

1. Adversarial Robustness (Lecture 3)

Main idea: an input is robust if small allowed changes do not change the prediction. Usually the perturbation set is an L_p ball around x.

Threat models to know

- $\|x' - x\|_\infty \le \epsilon$: each coordinate can move a little.
- $\|x' - x\|_2 \le \epsilon$: total Euclidean move is bounded.
- $\|x' - x\|_1 \le \epsilon$: sparse move, usually one or few coordinates matter.

Adversarial objective

$$\max_{\delta \in \Delta} L(x + \delta, y)$$

Take the perturbation that makes the loss as large as possible.

FGSM (must memorize) Linearize the loss around x:

$$L(x') \approx L(x) + \nabla_x L(x)^\top (x' - x)$$

For L_∞ constraint,

$$x^{adv,\infty} = x + \epsilon \,\mathrm{sign}(\nabla_x L(x))$$

Why: under an L_∞ bound, each coordinate should move to the edge in the sign of the gradient.

PGD intuition Repeat: take a gradient step to increase loss, then project back to the allowed set.

$$\delta \leftarrow \Pi_\Delta(\delta + \alpha \nabla_\delta L(x + \delta, y))$$

Think: FGSM is one step, PGD is many stronger steps.

General L_p attack from HW1

$$x^{adv,p} = x + \epsilon\, \phi_p(\nabla_x L(x)), \qquad \phi_p(z) = \arg \max_{z' \in B_p(0,1)} z^\top z'$$

Closed forms:

- $\phi_\infty(z) = \mathrm{sign}(z)$
- $\phi_2(z) = z/\|z\|_2$
- $\phi_1(z) = e_{i_{\max}(z)} \,\mathrm{sign}(z_{i_{\max}(z)})$

where $i_{\max}(z) = \arg\max_i |z_i|$.

Easy interpretation

- L_∞: perturb all features a little.
- L_2: move in the gradient direction.
- L_1: perturb the single coordinate with biggest gradient magnitude.

Very testable fact For the HW1 L_1 attack on a usual d-dimensional input, only **one component** is perturbed.

Targeted vs untargeted

- Untargeted: just cause any mistake.
- Targeted: force the model to predict a chosen label.

Common trap Certified robustness for L_2 at radius ϵ does **not** imply certified robustness for L_∞ at the same radius.

2. Jailbreaking LLMs (Lecture 4)

Main idea: make a language model ignore its safety rules by carefully changing the prompt.

Attacks to recognize

- GCG = Greedy Coordinate Gradient. White-box, token-level, optimized suffix attack.
- PAIR / TAP = automated black-box search attacks.
- Adaptive reasoning = uses the model's own reasoning to get around safety.

SmoothLLM Defense idea: make many slightly perturbed copies of the prompt (small character-level changes), run the model, then aggregate. If a jailbreak is brittle, tiny edits break it.

Most testable fact SmoothLLM is especially effective against **GCG**-style attacks.

3. Robust Training + Randomized Smoothing (Lecture 5)

Adversarial training Train on worst-case perturbations:

$$\min_\theta \mathbb{E}_{(x,y)} \left[\max_{\delta \in \Delta} L(x + \delta, y; \theta) \right]$$

Inner max finds a strong adversarial example. Outer min learns parameters that do well even on those examples.

Danskin theorem intuition If δ^* solves the inner max, then when taking gradient wrt θ, treat δ^* as the chosen worst-case perturbation:

$$\nabla_\theta \max_{\delta \in \Delta} L(x + \delta, y; \theta) = \nabla_\theta L(x + \delta^*, y; \theta)$$

So in practice: attack first, then backprop on attacked data.

Randomized smoothing Base classifier f, Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2 I)$:

$$g(x) = \arg\max_c \Pr(f(x + \eta) = c)$$

Let

$$p_A = \Pr(f(x + \eta) = A), \qquad p_B = \max_{B \ne A} \Pr(f(x + \eta) = B)$$

The bigger the gap between top class and runner-up, the better the certificate.

Radius formula to remember

$$R = \frac{\sigma}{2} \left(\Phi^{-1}(p_A) - \Phi^{-1}(p_B) \right)$$

Larger p_A, smaller p_B, or larger σ can help the radius.

Quick exam logic If choice A has $(p_A, p_B) = (0.60, 0.30)$ and choice B has $(0.52, 0.45)$, then A has larger certified radius because the top-two separation is much bigger.

4. Formal Verification (Lecture 6)

Big picture: turn network behavior into logic constraints and ask if a bad counterexample exists.

Formula to know

$$\phi_I(x, x_0) \wedge \phi_N(x_0, y_0) \wedge \phi_N(x, y) \wedge \neg(y = y_0)$$

where:

- ϕ_I: allowed perturbation set around x_0
- ϕ_N: network computation constraints
- $\neg(y = y_0)$: output changed

Meaning of satisfiable vs unsat

- SAT: there exists a perturbed input causing a different prediction, so the network is **not robust** at x_0.
- UNSAT: no such counterexample in that region, so robustness is verified there.

Terms

- SAT: satisfiability of Boolean formulas.
- SMT: SAT plus richer math constraints.
- Reluplex: SMT-style solver specialized for networks with ReLUs.
- Soundness: if solver says SAT/UNSAT, the result is correct.
- Completeness: solver can in principle finish with a definite answer.

5. Label Shift (Lecture 7)

Definition

$$p(x \mid y) = q(x \mid y), \qquad p(y) \neq q(y)$$

The class-conditional feature distribution stays the same, but class frequencies change.

Examples to remember

- Bird classification: migration changes species rates $\Rightarrow$ label shift.
- Text model: more of certain phrases or label types can be treated as label shift.

Most important invariance Under label shift,

$$\Pr_P(f(X) = i \mid Y = j) = \Pr_Q(f(X) = i \mid Y = j)$$

This is the key reason the confusion behavior conditioned on the true class stays stable.

Matrix view Define

$$C_{ij} = \Pr_P(f(X) = i, Y = j)$$

and class-weight vector

$$w_j = \frac{\Pr_Q(Y = j)}{\Pr_P(Y = j)}$$

Then predicted-label probabilities under the test distribution satisfy

$$q = Cw$$

This lets us estimate new class proportions from old data plus observed predictions.

Practice-question reminder If asked which statement is true under label shift, the safe answer is usually the conditional invariance above.

6. Covariate Shift (Lecture 8)

Definition

$$p(y \mid x) = q(y \mid x), \qquad p(x) \neq q(x)$$

The labeling rule is unchanged, but the input distribution changes.

Examples

- New microphone for speech data $\Rightarrow$ covariate shift.
- More snowy weather in bird images $\Rightarrow$ covariate shift.

Importance weighting

$$w(x) = \frac{q(x)}{p(x)}$$

Use this to reweight source/calibration samples so they look more like target/test samples.

Discriminator idea Train a binary classifier to distinguish source from target. Its output can help estimate density ratios and therefore importance weights.

What a bounded-Lipschitz discriminator is useful for

- computing importance weights under covariate shift
- estimating Wasserstein distance

Not the main tool for label-shift weights.

Distance ideas

- Total variation: based on all bounded test functions.
- Wasserstein: based on 1-Lipschitz test functions.

7. Conformal / PAC Prediction Sets (Lecture 9)

This is a likely long-answer topic.

Setup Use a score function $f(y \mid x)$. Higher score = label y looks more plausible for x. Prediction set at threshold τ:

$$\tilde{f}_\tau(x) = \{y : f(y \mid x) \geq \tau\}$$

Coverage goal Approximately correct at error ϵ means

$$\Pr(y^* \in \tilde{f}_\tau(x)) \geq 1 - \epsilon$$

A PAC guarantee says: with probability at least $1 - \delta$ over the calibration sample, the learned set achieves this coverage target.

Optimal threshold Let the calibration score random variable be $Z = f(y^* \mid x)$ with CDF F. Then

$$\tau^*(\epsilon) = F^{-1}(\epsilon)$$

Since inverse CDFs are increasing, $\tau^*(\epsilon)$ is an **increasing** function of ϵ.

Inputs to PAC conformal algorithm

- score function $f(y \mid x)$
- held-out calibration data Z
- PAC parameters (ϵ, δ)

Not an input: the threshold τ itself, because the algorithm chooses it.

Regression case (easy points) If

$$f(y \mid x) = |y - g_\theta(x)|$$

then the prediction set is the interval

$$\tilde{f}_\tau(x) = [g_\theta(x) - \tau,\ g_\theta(x) + \tau]$$

Length of the interval $= 2\tau$.

Covariate-shift PAC result from HW2 If $w(x) = q(x)/p(x)$ and $w(x) \leq w_{\max}$ for all x, then

$$F_Q(\alpha) \leq w_{\max} F_P(\alpha)$$

which implies

$$\tau_Q^*(\epsilon) \geq \tau_P^*(\epsilon / w_{\max})$$

Practical meaning: if calibrating on P but want guarantee on Q, using $\epsilon / w_{\max}$ on the source side is conservative and safe.

Rejection sampling idea from HW2 Accept a source sample with probability $w(x)/w_{\max}$. After resampling until acceptance, accepted samples behave like target samples from Q. Then run ordinary PAC prediction-set calibration on the accepted set.

8. Calibration + ECE (Lecture 10)

Perfect calibration If the model predicts confidence p, then among all such predictions, it should be correct about a fraction p of the time:

$$\Pr(Y = \hat{Y}(X) \mid \hat{p}(X) = p) = p$$

Binned calibration Group predictions into bins B_i by confidence.

$$\text{Accuracy}(B_i) = \text{fraction correct in bin } i$$
$$\text{Confidence}(B_i) = \text{average predicted confidence in bin } i$$

Perfect calibration in each bin means

$$\text{Accuracy}(B_i) = \text{Confidence}(B_i)$$

Empirical ECE formula

$$\text{ECE} = \sum_{i=1}^{k} \frac{|B_i|}{|Z|} \left|\text{Accuracy}(B_i) - \text{Confidence}(B_i)\right|$$

Smaller is better. Zero means perfectly calibrated on that dataset under that binning.

Histogram binning Replace all confidences in a bin by that bin's empirical accuracy.

Very testable fact If each bin has exactly **one calibration example**, then empirical ECE on that same calibration set is **0**. Why: each bin's confidence equals its accuracy by construction.

Calibration under covariate shift Even if $p(y \mid x) = q(y \mid x)$, calibration may **not** be preserved after shifting from $p(x)$ to $q(x)$. So "same conditional labels" does not automatically mean "same calibration." This is easy to miss.

9. Rapid-Fire Facts / Last-Minute Memory Dump

- FGSM: $x + \epsilon\, \text{sign}(\nabla_x L)$
- PGD = repeated gradient ascent + projection
- L_2 attack direction: normalized gradient
- L_1 attack direction: one best coordinate only
- SmoothLLM best known for defending against GCG
- SAT in verification = found counterexample = not robust
- Label shift: $p(x \mid y) = q(x \mid y)$
- Covariate shift: $p(y \mid x) = q(y \mid x)$
- Conformal set: $\{y : f(y \mid x) \geq \tau\}$
- $\tau^*(\epsilon) = F^{-1}(\epsilon)$ and is increasing in ϵ
- Regression conformal interval: $[g_\theta(x) - \tau,\ g_\theta(x) + \tau]$
- Perfect calibration means accuracy matches confidence
- One sample per histogram bin $\Rightarrow$ empirical ECE $= 0$

10. What I would write for partial credit if I blank out

- **Robustness question:** define the perturbation set, say attack maximizes loss in that set, and mention FGSM/PGD.
- **Formal verification question:** write the SAT formula idea and say SAT means counterexample exists.
- **Label vs covariate shift:** first write which conditional stays the same.
- **Conformal question:** define prediction set by thresholding score function and say coverage should be at least $1 - \epsilon$.
- **Calibration question:** define confidence vs accuracy in a bin, then write ECE.